

Ega Hemachandra

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SKILLS

- **Programming & Markup Languages:** Python, C, C++, Java, HTML, CSS, JavaScript
- **Frameworks & Libraries:** Pandas, Numpy, Matplotlib, Mesa, NetworkX
- **Tools & Platforms:** MySQL, Git, GitHub, VS Code, Docker, Jupyter Notebook
- **Core CS Fundamentals:** DSA, OOPs, DBMS, Operating Systems, Computer Networks, Machine Learning
- **Soft Skills:** Problem-Solving Skills, Adaptability, Team Collaboration, Attention to Detail, Critical Thinking

PROJECTS

AI-based music genre classification (ML Project) | [GitHub](#)

Nov 2025 – Dec 2025

- Classified 7 music genres using an SVM model trained on GTZAN dataset. Achieved 73% test accuracy using MFCC and spectral audio features.
- Built an interactive Streamlit web application enabling users to upload audio, predict genres, and visualize results with a dynamic 10-band equalizer UI
- Designed a prototype adaptive equalizer system that automatically adjusts frequency bands based on predicted genre to enhance user listening experience
- Tech Stack: Python, Streamlit, Scikit-learn, Librosa, NumPy

AI-Powered RTI Query Classification System | [GitHub](#)

Nov 2025 – Dec 2025

- Built an AI model to classify user queries into sectors (Education, Healthcare, Infrastructure).
- Implemented AI-assisted responses using a knowledge base with keyword matching.
- Designed a human-in-the-loop admin dashboard to review, reclassify, and respond to queries.
- Tech Stack: Python, Flask, SQLite, Scikit-learn, HTML/CSS, JavaScript

TRAINING

• Fundamentals of Data Structures: Learn, Apply and Build Projects | [Lpu](#)

Jan 2025 – Aug 2025

- Learned core data structures such as arrays, linked lists, stacks, queues, trees, graphs, and hash tables, along with their real-world applications.
- Analyzed time and space complexity to understand performance trade-offs and choose the right data structure for given problems.
- Implemented data structures in code through hands-on practice, strengthening understanding of internal working mechanisms.

CERTIFICATES

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|---|----------|
| • Computer Networks Coursera | Oct 2024 |
| • Hack-A-Thron AIESEC | Aug 2024 |
| • Web-A-Thon Student Welfare Wing - Lpu | Sep 2024 |

Extra Curricular Activities

Senior Officer – Student Career Committee, SCSE, LPU

Mar 2026 – Present

- Organized expert sessions on Open Source and GSoC, Leaders Talk, and Tech Fluence 6; coordinated with mentors and teams to ensure smooth event execution and participant engagement.
- Managed event coordination, maintained student records using Google Sheets, and supported the organization of technical sessions and workshops.
- Built an automation bot for data extraction and entry, reducing 4 days of manual work to under 2 hours

EDUCATION

Lovely Professional University

Bachelor of Technology (B.Tech) – Computer Science and Engineering,

Phagwara, Punjab

Aug 2024 – Present

Geethanjali Institute of Science And Technology

Diploma - 69.9%

Nellore, Andhra Pradesh

Aug 2021 – June 2024

Kendriya Vidyalaya, Nellore

Matriculation - 56%

Nellore, Andhra Pradesh

Apr 2020 – Mar 2021